

CREATININE FACTORY

COCKROFT-GAULT

$$\frac{(140 - \text{age}) \times \text{wt}}{72 \times \text{PCr} \times 0.85}$$

DRUG DOSING
MDRD = BETTER <60

TRUE GFR <30
~~**eGFR 65**~~
✓ NORMAL

CONSTANT RATE
MUSCLE
SARCOPENIA
FILTERED FREELY

COCKROFT-GAULT
MDRD
NOT FOOLED BY MUSCLE
BETTER BELOW 60
CKD-EPI MOST ACCURATE OVERALL
DRUG DOSING

STEADY STATE REQUIRED

AKI - CREATININE RISING DAILY

FORMULAS INVALID
DO NOT USE FOR DRUG DOSING

FALSE LOW GFR
NOT FOOLED BY MUSCLE
CREATININE PELLETS
MUSCLE
CONSTANT RATE
FILTERED FREELY

1. Creatinine produced at constant rate

WHAT YOU SEE

'CONSTANT RATE' hopper feeding the line

WHAT IT ENCODES

Creatinine is generated at a steady rate from muscle creatine breakdown and is freely filtered with minimal tubular handling, making it a usable filtration marker — but only at steady state.

BOARD TRAP

Trusting serum Cr in non-steady-state AKI — when Cr is rising daily, no equation gives a valid GFR.

2. Muscle mass drives production

WHAT YOU SEE

Muscular figure labeled 'MUSCLE'

WHAT IT ENCODES

Creatinine generation is proportional to muscle mass. High muscle mass raises serum Cr independent of GFR; low muscle mass lowers it. This is the key confounder of all creatinine-based estimates.

BOARD TRAP

Reading a muscular patient's higher Cr as kidney disease — their GFR may be normal; their muscle mass is the driver.

3. Sarcopenia → false-normal Cr

WHAT YOU SEE

Frail elderly woman 'SARCOPENIA'

WHAT IT ENCODES

Low muscle mass (elderly, sarcopenia, cachexia, amputees) produces little creatinine, so a 'normal' Cr can hide a markedly reduced GFR. Use cystatin C or measured clearance, and adjust drug dosing carefully.

BOARD TRAP

Reassuring a frail elderly patient with 'normal' Cr — their true GFR may be severely reduced (false-low Cr scenario).

4. Filtered freely

WHAT YOU SEE

'FILTERED FREELY' conveyor

WHAT IT ENCODES

Creatinine is filtered freely at the glomerulus but a small amount is also secreted by proximal tubules, causing creatinine clearance to slightly overestimate true GFR. Drugs like trimethoprim/cimetidine block secretion and raise Cr without lowering GFR.

BOARD TRAP

Interpreting a Cr bump on trimethoprim/cimetidine as true GFR decline — they block tubular secretion, not filtration.

5. True GFR <30 needs caution

WHAT YOU SEE

'TRUE GFR <30' crossed over eGFR 65

WHAT IT ENCODES

At low GFR, tubular secretion contributes proportionally more, so eGFR/creatinine clearance can substantially overestimate true GFR (an 'eGFR 65' may mask true GFR <30). Confirm advanced CKD with cystatin C or measured clearance.

BOARD TRAP

Trusting eGFR near the low range as accurate — it overestimates true GFR most when function is poor.

6. Cockcroft-Gault: drug dosing

WHAT YOU SEE

'COCKCROFT-GAULT — DRUG DOSING' gauge

WHAT IT ENCODES

Cockcroft-Gault = [(140–age) x weight x 0.85 if female] / (72 x SCr); estimates creatinine clearance (mL/min). It is the equation most drug package inserts are validated against, so use it for renal drug dosing.

BOARD TRAP

Dosing nephrotoxic/renally-cleared drugs off CKD-EPI eGFR — package-insert thresholds are based on Cockcroft-Gault.

7. MDRD better below 60

WHAT YOU SEE

'MDRD' gauge reading '<60'

WHAT IT ENCODES

The MDRD equation was derived in CKD populations and performs best when eGFR is <60. It is reported as mL/min/1.73m2 (indexed to BSA) and underestimates at higher GFRs.

BOARD TRAP

Using MDRD to label mild reductions — it is unreliable and biased above 60; it shines below 60.

8. CKD-EPI most accurate overall

WHAT YOU SEE

'CKD-EPI MOST ACCURATE OVERALL' bottle

WHAT IT ENCODES

CKD-EPI is more accurate than MDRD across the GFR range, especially above 60, and is the preferred staging equation. Cystatin C–based CKD-EPI further reduces muscle-mass bias.

BOARD TRAP

Defaulting to MDRD for staging — CKD-EPI is the recommended, more accurate equation across the spectrum.

9. Steady state required

WHAT YOU SEE

'STEADY STATE REQUIRED' green light

WHAT IT ENCODES

All creatinine equations assume steady-state kinetics. They are invalid in rapidly changing function (AKI), pregnancy, extremes of body size, or unusual diets. Use only when Cr is stable.

BOARD TRAP

Plugging a rapidly rising Cr into an eGFR formula — the result is meaningless in non-steady-state AKI.

10. AKI — creatinine rising daily

WHAT YOU SEE

'AKI — CREATININE RISING DAILY' red alarm

WHAT IT ENCODES

In AKI the serum creatinine lags behind the true fall in GFR and changes daily, so any eGFR is unreliable. Manage by the creatinine trend and clinical context, not a calculated GFR.

BOARD TRAP

Quoting an 'eGFR' during active AKI — the number underestimates how low GFR truly is and shouldn't guide dosing.

11. Formulas invalid in non-steady state

WHAT YOU SEE

'FORMULAS INVALID' trash can

WHAT IT ENCODES

When steady-state is violated (AKI, dialysis, acute illness, large fluid shifts) estimating equations and even drug-dosing formulas break down — discard them and rely on measured clearance or clinical judgment.

BOARD TRAP

Continuing to dose by formula during acute changes — switch to clinical judgment / measured clearance instead.

12. Not fooled by muscle (drug-dosing worker)

WHAT YOU SEE

Worker 'NOT FOOLED BY MUSCLE'

WHAT IT ENCODES

Cystatin C is produced by all nucleated cells and is largely independent of muscle mass, so cystatin C–based eGFR is less confounded by sarcopenia or high muscularity. Use it to confirm when creatinine is unreliable.

BOARD TRAP

Relying solely on creatinine in extremes of muscle mass — cystatin C cross-checks and removes the muscle bias.
